Wheel-Ski Indication (Mod S.O.O. 6001) – General data

1. General

Refer to Figure 1 for details.

Wheel-ski position indication is provided by a group of six lights mounted on the selector lever panel (refer to 32-30-00, Extension and Retraction (Wheel-Ski) (Mod S.O.O. 6001) – General Data) which are controlled by limit switches on each ski assembly. The three upper lights indicate skis "up" and the lower three skis "down". Each pair of lights are located on the panel in relation to their respective ski position (for example, the two center lights provide nose ski indication). Two limit switches installed on each ski are actuated by the ski latch lever, one when the ski is "down" and the other when the ski is "up". The power supply for the indicator lights is derived from the 28 Volt left DC busbar, through a 5-ampere GEAR IND (H7) circuit breaker located on the avionics circuit breaker panel, and a relay operated by the CAUTION LIGHTS DIM/BRIGHT/TEST switch, refer to 33-10-11, Caution Lights Dimming Control Box – General Data. When the skis are operated, the relevant position limit switches are actuated to provide a ground and complete the electrical circuit for their associated lights. When the CAUTION LIGHT dimmer switch is selected to DIM, the relay changes over to connect a resistance in series with the lights, which are connected in parallel. For electrical schematic refer to PSM 1-64-2W, Section 32-61-00.

2. Adjustment/Test

Note The access panels from part of the stressed structure of the ski and must be reinstalled as soon as adjustments are completed.

A. Adjust Ski Up Limit Switch

Refer to Figure 1 for details.

1) Remove forward access panel. Check that ski is in ski-up position and that latch pin is fully extended and latched.

2) Loosen up-lock microswitch mounting screws.

3) Move switch toward latch pin until an audible click indicates that switch is actuated. Move switch an additional 1/32 inch approximately (to provide overtravel for consistent switch actuation in service) but ensure that latch pin does not come in direct contact with switch body.

4) Tighten switch mounting screws.

5) Using continuity tester, function check switch actuation by operating the ski hydraulically in and out of lock. If ski is installed on aircraft, check indicator lights in flight compartment. Ski UP/LAND light should come on when latch pin engages locking hole and light should go out when latch pin comes out of lock.

6) Reinstall access panel.

B. Adjust Ski Down Limit Switch

Refer to Figure 1 for details.

1) Remove forward access panel.

2) Loosen microswitch mounting screws.

3) Actuate ski to ski down position.

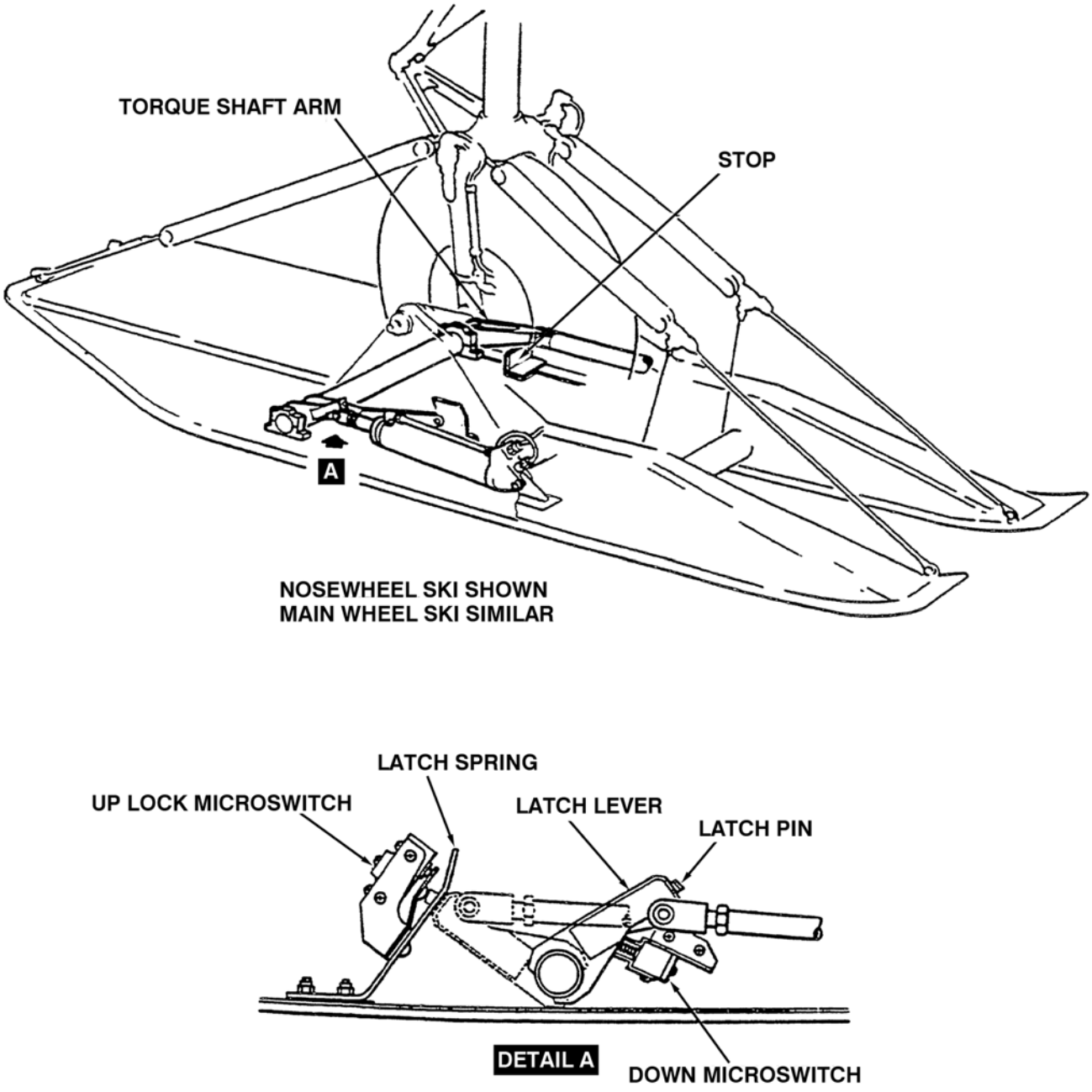
4) Move switch and actuator toward latch lever arm until an audible click indicates that switch is actuated.

5) Final adjustment of switch must result in actuation of switch when torque shaft arms are within 0.2 inch of their stops.

CAUTION **FAILURE TO COMPLY WITH THE 0.2 INCH ADJUSTMENT REQUIREMENT COULD RESULT IN PREMATURE OPERATION OF SKI POSITION INDICATOR LIGHT (BEFORE SKI HAS REACHED THE GEOMETRIC LOCK).**

6) Function check with continuity tester if skis are off aircraft. Function check by means of indicator lights in flight compartment if skis are on aircraft.

7) Reinstall access panel.



Limit Switch Adjustment
Figure 1